

Ekaagra Gupta

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🌐 Portfolio — 🐙 GitHub — 🔗 LinkedIn

Education

Manipal University Jaipur 2024 – 2028
Bachelor of Technology in Computer Science and Engineering Specialization in Artificial Intelligence and Machine Learning
Gyan Ashram School Batch 2024
Senior Secondary (Class XII) **Warren Academy School** Batch 2022
Secondary (Class X)

Technical Skills

Artificial Intelligence & Machine Learning: Machine Learning, Deep Learning, Computer Vision, XAI

Generative AI & Retrieval Systems: LLMs, RAG, LangChain, LlamaIndex, Hugging Face Transformers, vLLM, FAISS, ChromaDB, BM25, ColBERT, HNSW

Geospatial AI & Image Processing: Satellite Image Processing, NDVI/NDWI Spectral Analysis, Geospatial Analysis, Remote Sensing Concepts

Programming Languages: Python, C++, C, SQL, javascript

Full-Stack Development: MongoDB, Express.js, React.js, Node.js (MERN Stack)

Libraries & Frameworks: PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib

Course-work: OS, OOPs, DSA, Computer network, DBMS

Tools & Platforms: Git, GitHub, Linux, Docker, Jupyter Notebook, Google Colab, CUDA

Internship Experience

AI Intern May 2026 - Present
Malaviya National Institute of Technology Jaipur (MNIT)

- Led a modular Retrieval-Augmented Generation (RAG) pipeline with hybrid dense and sparse retrieval.
- Implemented document chunking, embedding generation, HNSW indexing, and BM25-based retrieval using Qwen embedding models.
- Integrated multi-stage reranking using Qwen Reranker, ColBERT, and Reciprocal Rank Fusion (RRF) for retrieval optimization.
- Optimized large-scale inference workflows using vLLM, Hugging Face Transformers, and PyTorch on NVIDIA GPUs.

Technologies: Python, PyTorch, Hugging Face Transformers, vLLM, Qwen Embeddings, ColBERT, BM25, HNSW, FAISS, RAG, Information Retrieval, Linux, Git, CUDA

AI/ML Intern Mar 2026 – May 2026
CoreGen

- Incorporated a **Super-Resolution Image Reconstruction** project for satellite imagery, building a multi-frame (3+1) deep learning pipeline that combines optical flow with Fourier-based neural warping to reconstruct high-resolution detail from low-resolution inputs ($64 \times 64 \rightarrow 256 \times 256$), with a custom GPU-optimized dataset and training pipeline for Earth observation applications.

- Operationalized a full-stack **MERN dashboard** (React frontend, Node/Express backend) to visualize and serve reconstruction outputs, connecting the model pipeline to a usable interface for reviewing results.
- Explored lightweight object detection (MobileNet SSD) and real-time anomaly detection for traffic analysis, evaluating ROS, TensorFlow Lite, and OpenCV for efficient on-device inference.

Technologies: Python, PyTorch, FAISS, ChromaDB, LangChain, LlamaIndex, HuggingFace Transformers, MobileNet SSD, ROS, TensorFlow Lite, OpenCV, MongoDB, Express.js, React.js, Node.js

Projects

MemoRAG: Memory-Augmented RAG Framework

 GitHub

LLM / RAG System

- Engineered a memory-augmented RAG pipeline that builds a compressed KV-cache representation of long documents before retrieval, enabling globally-aware query rewriting and surrogate-question generation for multi-hop and summarization queries that standard chunk-based RAG handles poorly.
- Integrated a dense retrieval layer using FAISS and BGE-M3 embeddings, paired with a lightweight variant (Qwen2.5-1.5B) for lower-resource deployment.
- Evaluated pipeline performance on long-context QA and summarization tasks, comparing standard RAG against the memory-augmented approach for retrieval accuracy and answer completeness.

Technologies: Python, HuggingFace Transformers, FAISS, LangChain, LLM APIs, Retrieval Systems

SEVAS: Satellite-based Environmental Violation Analysis System

 GitHub

AI/ML System

- Orchestrated a satellite image analysis system combining a custom U-Net segmentation model with **NDVI/NDWI spectral indices** to detect sand mining, land encroachment, and vegetation loss; U-Net achieved 73% validation on a 4-class pixel segmentation task (mining, vegetation, water, background).
- Leveraged Gemini Vision API for natural-language violation descriptions alongside the segmentation pipeline, combining computer vision with vision-language analysis for richer detection output.
- Designed a severity-scoring pipeline to prioritize violation reports for field inspection, reducing manual triage effort through automated ranking.
- Devised a Flask REST API supporting single and batch satellite image analysis (up to 50MB per file) across JPG, PNG, TIFF, and GeoTIFF formats.

Technologies: Python, TensorFlow, Keras, OpenCV, NumPy, scikit-learn, Flask, REST APIs, Geospatial Analysis, Gemini Vision API

GETHER: Generative Emission Temporal Hybrid Explainable Regressor



GitHub

AI/ML System

- Formulated a temporal deep learning pipeline for AQI forecasting using a stacked LSTM architecture (128→64 units) with engineered lag and rolling-statistic features, benchmarked against linear regression and random forest baselines.
- Engineered a causal discovery module using Granger causality and dynamic causal graph construction to surface relationships between pollutants (PM2.5, NO2, SO2) and AQI

variations.

- crafted a SHAP-based explainability layer to rank pollutant feature importance and interpret individual model predictions.
- Configured a counterfactual policy simulator to estimate AQI impact under hypothetical emission-reduction scenarios, deployed via an interactive Streamlit dashboard.

Technologies: Python, TensorFlow, Scikit-learn, SHAP, Streamlit, Pandas, NumPy, Matplotlib

Patent application filed (under examination)

Leadership and Achievements

Google Developer Groups — Technical Co-Lead May 2026 – Present

- Led technical planning and execution of developer-focused workshops on web, cloud, and AI/ML technologies — *300+ attendee / 3 workshops* .

IEEE CIS — Joint Head, Sponsorship and Curation Jan 2026 – Present

- Led sponsorship acquisition for technical events — *150+ sponsors secured* and contributed to the webmaster team.

DevForge — Technical Joint Head Jul 2026 – Present

- Oversaw full-stack development across *24+* projects, focusing on modular architecture and performance optimization.

GSSoC'25 — Open Source Contributor Jul 2025

- Contributed AI/ML and web features across *90+* merged pull requests following structured open-source workflows.

Certifications

- **Prompt Engineering & Programming with OpenAI** — Columbia University
Credential Link
- **Essentials in Generative AI** — Microsoft & LinkedIn
Credential Link
- **Generative AI Studio** — Google Cloud
Credential Link
- **Generative AI: The Evolution of Thoughtful Online Search** — LinkedIn Learning
Credential Link

Publications

- **Mapping the Latency-Explainability-Accuracy Pareto Frontier: A Systematic Benchmark and Framework for Attribution Methods in Real-Time Satellite Anomaly Detection**
Manuscript in preparation, targeting IEEE Transactions on Geoscience and Remote Sensing